

decoder. IC4 converts the 12-bit data into 8-bit address and 4-bit data. DIP2 is used to set the address of the decoder.

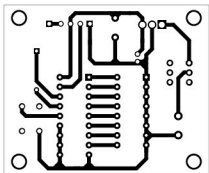


Fig. 3: Actual-size PCB layout of the transmitter unit

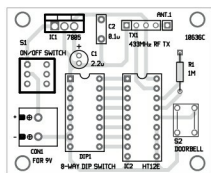


Fig. 4: Component layout of the PCB shown in Fig. 3

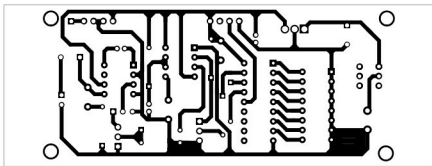


Fig. 5: Actual-size PCB layout of the receiver unit

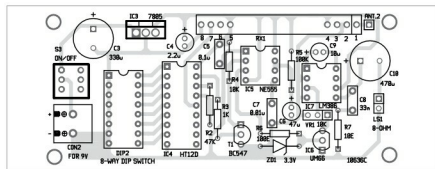


Fig. 6: Component layout of the PCB shown in Fig. 5

The 8-bit address of the decoder must match with the encoder to receive the information. The decoder checks the serial input three times continuously. If address bits of transmitter and receiver match, data is decoded and valid transmission VT pin of IC4 goes high. This triggers NE555 configured in monostable mode.

NE555 generates a high pulse for about five seconds, whose period is determined by resistor R5 and capacitor C6. Time period of NE555 is determined by the relationship:

$$\text{Time period (in seconds)} = 1.1 \times R5 \times C6$$

It means, when S2 is pressed momentarily, provided S1 and S3 are closed, output pin 3 of IC5 goes high for about five seconds. This output pulse activates the melody generator (IC6) and so the melody sounds up to about five seconds.

Zener diode ZD1 regulates the output of IC5 to 3.3V, which drives IC6. Output of IC6 is given to IC7 through potmeter VR1. Gain of the audio amplifier is set to 200. VR1 controls the volume of the sound before amplification.

PARTS LIST

Semiconductors:

- IC1, IC3 - 7805, 5V voltage regulator
- IC2 - HT12E encoder
- IC4 - HT12D decoder
- IC5 - NE555 timer
- IC6 - UM66/BT66 musical melody
- IC7 - LM386 audio amplifier
- T1 - BC547 npn transistor
- ZD1 - 3.3V Zener diode

Resistors (all 1/4-watt, 45% carbon):

- R1 - 1-mega-ohm
- R2 - 47-kilo-ohm
- R3 - 1-kilo-ohm
- R4 - 10-kilo-ohm
- R5 - 100-kilo-ohm
- R6 - 100-ohm
- R7 - 10-ohm
- VR1 - 10-kilo-ohm potmeter

Capacitors:

- C1, C4 - 2.2μF, 16V electrolytic
- C2, C5 - 0.1μF ceramic disk
- C3 - 330μF, 25V electrolytic
- C10 - 470μF, 25V electrolytic
- C6 - 47μF, 16V electrolytic
- C7 - 0.01μF ceramic disk
- C8 - 33nF ceramic disk
- C9 - 10μF, 16V electrolytic

Miscellaneous:

- CON1, CON2 - 2-pin terminal connector
- S1, S3 - On/off switch
- S2 - Pushbutton switch
- TX1 - 433MHz RF transmitter
- RX1 - 433MHz RF receiver
- LS1 - 8-ohm, 0.5W loudspeaker
- ANT1, ANT2 - 17cm single-strand hook-up wire antenna
- 9V - 9V battery or 9V DC power supply

Construction and testing

An actual-size, single-side PCB layout for the transmitter is shown in Fig. 3 and its component layout in Fig. 4. Similarly, an actual-size, single-side PCB layout for the receiver is shown in Fig. 5 and its component layout in Fig. 6.

After assembling the circuits on two separate PCBs, enclose these in suitable plastic boxes. Use approximately 17cm long single-strand hook-up wire antenna for the transmitter and the receiver each. The receiver unit requires a well-regulated 9V DC power supply for low noise and efficient operation. Alternatively, a 9V battery each for the transmitter and the receiver may be used as power supplies. **BY**



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